Energy cooperatives

CleantechHUB as a partner against energy poverty

Embassy meeting · 11 Nov 2026 · 20 min

The national gap

15,4%
IMPE 2024 · ~8.1 M people

9,5%
2030 scenario · ~5.17 M (−2.9 M)

75,92% / 93,12%
ICEE rural vs national (2024)

63,1% / 52,8%
IMPE La Guajira / Chocó (2024)

Lifting ~2.9 million people out of energy poverty is the gap between the 2024 baseline and a single-digit incidence scenario in 2030.

Source: IMPE 2025, 3rd edition (2024 base year). · UPME ICEE Bulletin 2024; 93.12% co (2024).

Pace toward 2030

~300 k people/yr
Observed pace
~2034

~490 k people/yr
IMPE 2030 scenario
2030

490 k × 6 years ≈ 2.94 M people · ≥1.0 pp/year incidence

Source: IMPE 2025, pp. 12, 33.

Not just CAPEX: the risk is OPEX

- **Contraloría:** concern over sustainability when operation and administration of microgrids and off-grid PV are not remunerated; maintenance and monitoring failures.
- **Amazonas:** ~$30,000 million (royalties) in PV not operating; damaged batteries — critical sustainability.
- **CREG:** thousands of rural PV systems without management → often abandoned; insufficient rules, certified installers, and O&M.

Sources: El Colombiano (Contraloría); Seguimiento.co/Procuraduría (Amazonas); CREG [illegible]5-13; CREG Concept 4022/2019 (O&M).

The cooperative as the tool

CTH PROPOSAL

Community O&M + revenue + local ownership complements CAPEX with OPEX contracts (PV, batteries, monitoring; CREG/SSPD) and community governance.

KPIs: IMPE access and quality — does not replace ICEE or IPEM.

Local talent and green jobs

- **Escuela Colombiana de Ingeniería Julio Garavito** — collaboration on training and green jobs.
- CLSA and energy-sector challenges; 3rd place Solucionatón 10X (2026).
- Renewable diploma/courses and 101.76 kWp laboratory.

Source: escuelaing.edu.co — CLSA, Solucionatón 10X, renewable lab.

Productive communities (PASO)

Productive and cooperative projects — PASO does not run energy projects.

- **Fonseca, La Guajira** — COOMPAZCOL, ASMUPERIJA (IMPE 63.1%)
- **Puerto Asís / Puerto Guzmán, Putumayo** (IMPE 58.7%)
- **Istmina, Chocó** — Bioconversión Chocó, Aserrío Continental (52.8%)
- **San José del Guaviare** (IMPE 38.9%)
- **Arauca** — COOPCACAO, COMPROCAR, Filimarpaz (27.9%)

Energy Coop candidate community: Corozal, Sucre (IMPE 43.1%) — not a PASO project.

Data lab: candidate funnel

~1.000 → 100 → 20 → 10

- Base: ~1,000 prioritised communities (MinEnergía).
- Filters: energy poverty, ZNI/off-grid (UPME), solar resource, legal ID/NIT, social licence.
- Candidates, not pilots — scores not published.

Source: CTH/PASO brief 2026; public IMPE/ZNI/UPME data.

Potential partner network

FMO
Solar finance at scale in Colombia

RVO
Facilitation and business cooperation

Green Choice
Finance for community energy

Energie Samen
Peer learning across cooperatives

Potential partners — no signed agreements at this stage.

Sources: fmo.nl (Colombia); CTH brief 2026 (Green Choice, Energie Samen).

Next steps

1. Working session with the full team and Juan Fernando Lucio (PASO Colombia).
2. Align the candidate funnel, cooperative role, and financing needs.
3. Contact: cleantechhub.net

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